higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

16 February 2022

This marking guideline consists of 7 pages.

MARKING GUIDELINE
NATED

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SECTION A

QUESTION 1

1.1 A
1.2 D
1.3 B
1.4 A
1.5 D

(5 × 2) **[10]**

QUESTION 2

2.1 False
2.2 False
2.3 True
2.4 True
2.5 False
2.6 False
2.7 True
2.8 True

(8 × 1) **[8]**

QUESTION 3

3.1 C
3.2 G
3.3 B
3.4 F
3.5 D
3.6 E
3.7 A

(7 × 1) **[7]**

TOTAL SECTION A: 25

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SECTION B

QUESTION 4

4.1

BINARY	1'S COMPLIMENT
0 • 0 0 1 0 0 1 , 0 1 0 0	0 • 0 0 1 0 0 1 , 0 1 0 0
1 • 0 1 0 0 0 0 , 0 1 0 1	1 • 1 0 1 1 1 1 , 1 0 1 0
	0 1 • 1 1 1 0 0 0 , 1 1 1 0
	0
	1 • 1 1 1 0 0 0 , 1 1 1 0
	NEG 1 1 1 , 0 0 0 1 $_2$

Allocate 1 mark for each of the following steps indicated correctly.

0 * 0 0 1 0 0 1 , 0 1 0 0 ✓

1 * 1 0 1 1 1 1 , 1 0 1 0 ✓

0 1 * 1 1 1 0 0 0 , 1 1 1 0 ✓

1 * 1 1 1 0 0 0 , 1 1 1 0 ✓

NEG 1 1 1 , 0 0 0 1 $_2$ ✓

4.2

2'S COMPLEMENT

0 * 0 1 0 0 1 , 0 1 0

1 * 0 0 1 0 0 , 0 1 1

0 * 0 1 0 0 1 , 0 1 0

1 * 1 1 0 1 1 , 1 0 1

1 0 * 0 0 1 0 0 , 1 1 1

POS 1 0 0 , 1 1 1 $_2$

Allocate 1 mark for each of the following steps indicated correctly.

0 * 0 1 0 0 1 , 0 1 0 ✓

1 * 1 1 0 1 1 , 1 0 1 ✓

1 0 * 0 0 1 0 0 , 1 1 1 ✓

POS ✓ 1 0 0 , 1 1 1 $_2$ ✓

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4.3

$$
\begin{array}{rrrrrrrrrrrrr}
 & & & & 1 & 0 & 0 & 1 & 0, & 0 & 1 & _2 \\
 & & & X & & 1 & 0 & 0 & 1, & 1 & 1 & _2 \\
\hline
 & & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & \\
 & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & \\
 & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & \\
 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & & \\
\hline
 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1, & 1 & 1 & 1 & 1 & _2
\end{array}
$$

Allocate 2 marks where it was indicated the following number must be added.

$$
\begin{array}{rrrrrrrrrrrrr}
 & & & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & \\
 & & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & ✓ \\
 & & & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & & \\
 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & & & & & & ✓ \\
\hline
\end{array}
$$

Allocate 2 marks if the answer is correct.

$$1\ 0\ 1\ 1\ 0\ 0\ 0\ 1,\ 1\ 1\ 1\ 1\ _2 \quad ✓✓$$

Allocate 1 mark if the comma is paced in the correct position.

$$1\ 0\ 1\ 1\ 0\ 0\ 0\ 1,\ 1\ 1\ 1\ 1\ _2 \quad ✓$$

4.4

$$
\begin{array}{rrrrrrrrrrrrrrrrrrrrr}
 & & & & & & & & & & & & & & 1 & 0 & 0 & 1 & 0, & 0 & 1 & _2 & ✓✓ \\
 & & & 1 & 0 & 0 & 1, & 1 & 1 & \multicolumn{1}{|r}{1} & 0 & 1 & 1 & 0 & 0 & 0 & 1, & 1 & 1 & 1 & 1 & & \\
\cline{10-20}
 & & & & & & & & & 1 & 0 & 0 & 1, & 1 & 1 & & & & & & & & ✓ \\
 & & & & & & & & & & & & & 1 & 0 & 1 & 0 & 1 & 1 & & & & \\
 & & & & & & & & & & & & & 1 & 0 & 0 & 1, & 1 & 1 & & & & ✓ \\
 & & & & & & & & & & & & & & & & 1 & 0 & 0 & 1 & 1 & 1 & \\
 & & & & & & & & & & & & & & & & 1 & 0 & 0 & 1, & 1 & 1 & \\
 & & & & & & & & & & & & & & & & & & & & & 0 & ✓
\end{array}
$$

Allocate 1 mark for each step as indicated.

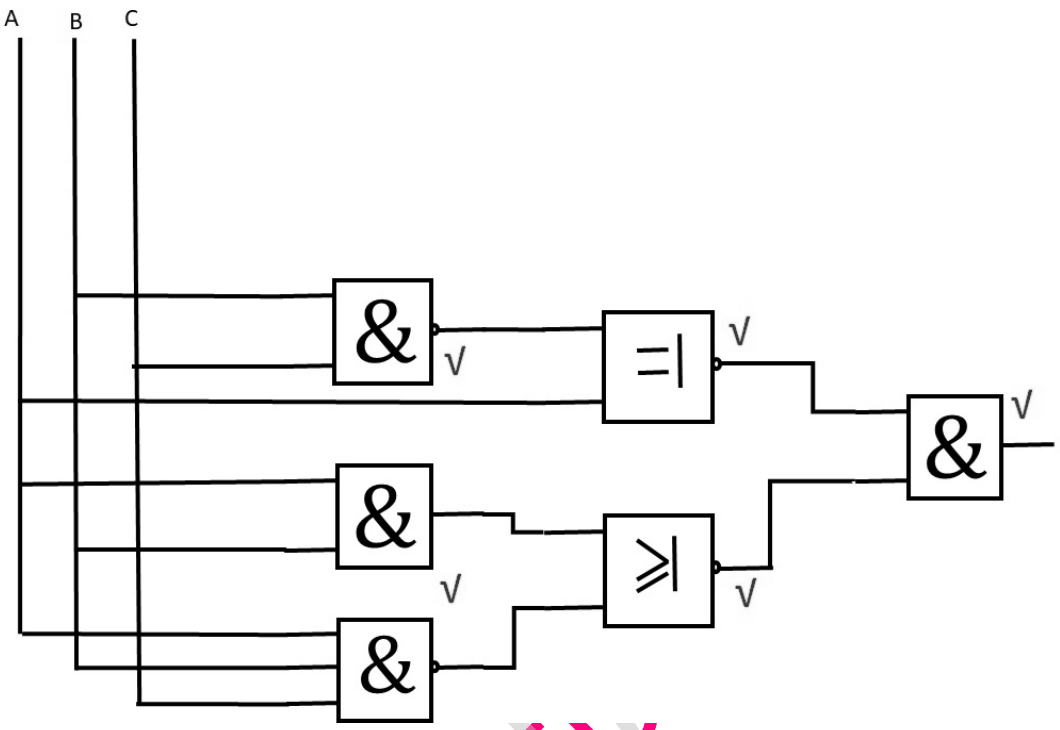
Allocate 1 mark if the comma was placed in the correct position.

(4 × 5) **[20]**

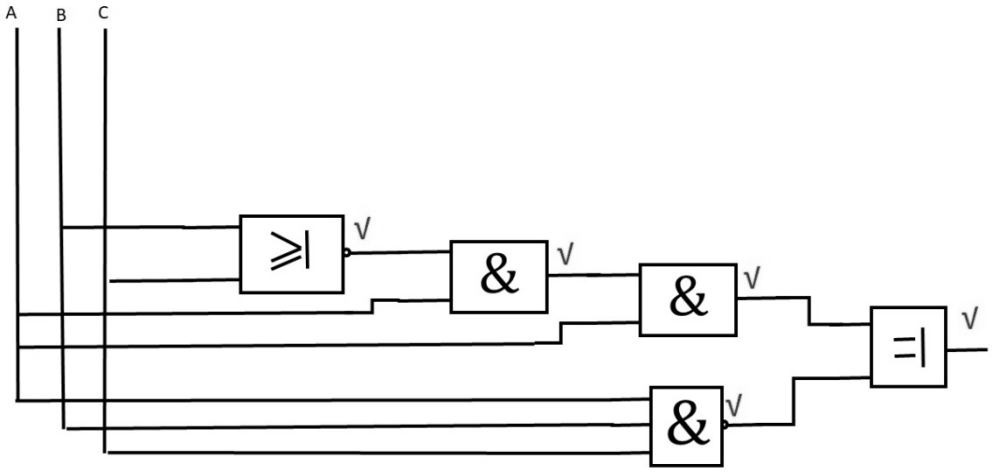
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QUESTION 5

5.1

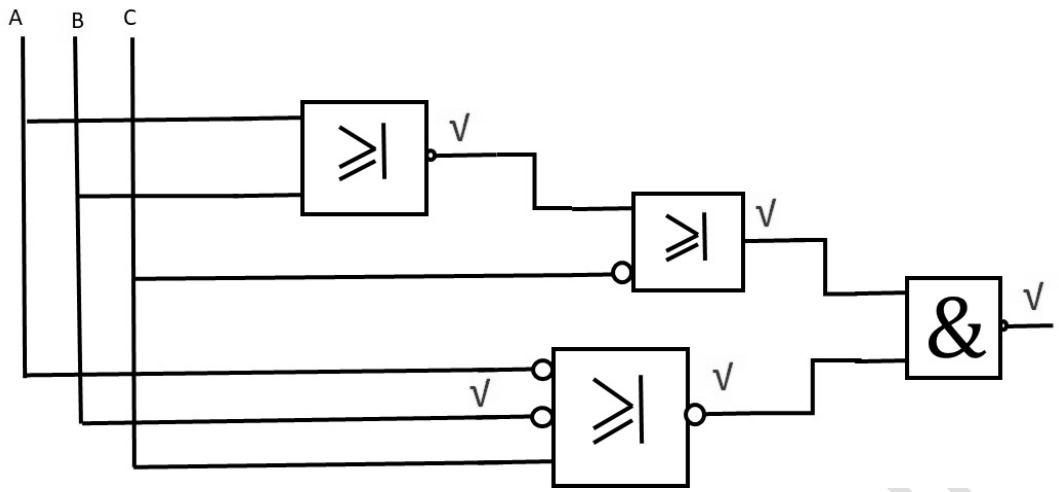


5.2



MA

5.3



(3 × 5) **[15]**

QUESTION 6

6.1 $\overline{\overline{(A+B\overline{C}+\overline{A}B)}+\overline{(\overline{A}+BC+A\overline{B})}}=f$

$=(A+B\overline{C}+\overline{A}B)\bullet(\overline{A}+BC+A\overline{B})$

$=A\overline{A}+ABC+AA\overline{B}+\overline{A}B\overline{C}+B\overline{C}BC+B\overline{C}A\overline{B}+\overline{A}\overline{A}B+\overline{A}BBC+\overline{A}BA\overline{B}$ ✓

$=0+ABC+A\overline{B}+\overline{A}B\overline{C}+0+0+\overline{A}B+\overline{A}BC+0$ ✓

$=ABC+A\overline{B}+\overline{A}B\overline{C}+\overline{A}B+\overline{A}BC$ ✓

$=A(BC+\overline{B})+\overline{A}B$ ✓

$=A(C+\overline{B})+\overline{A}B$

$=AC+A\overline{B}+\overline{A}B$ ✓

6.2 $\overline{\overline{(A\bullet\overline{B}+\overline{A}\bullet\overline{B}\bullet C)(\overline{A}\bullet\overline{B}\bullet C+\overline{A})}(\overline{\overline{A}\bullet\overline{B}\bullet C+A\bullet\overline{B}})}=f$

$=((A\overline{B}+\overline{A}\overline{B}C)\bullet(\overline{A}\overline{B}C+\overline{A}))+(\overline{A}\overline{B}C+A\overline{B})$ ✓

$=(\overline{A}\overline{B}C+A\overline{B})$ ✓

$=\overline{B}(\overline{A}C+A)$ ✓

$=\overline{B}(C+A)$ ✓

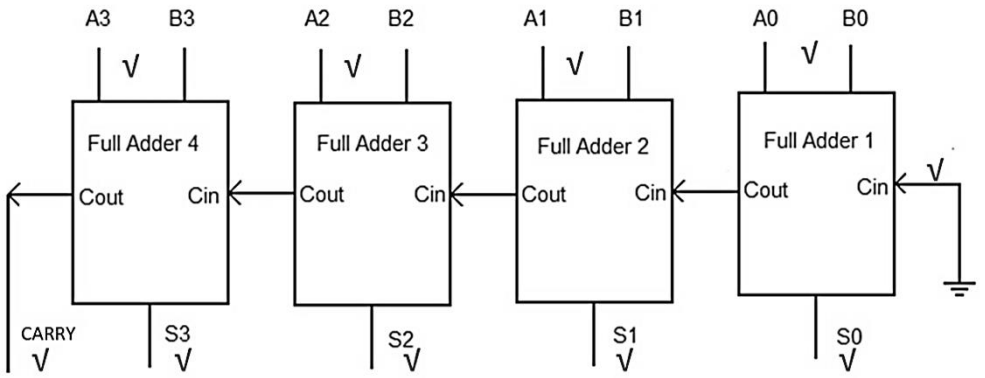
$=\overline{B}C+A\overline{B}$ ✓

(2 × 5) **[10]**

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QUESTION 7

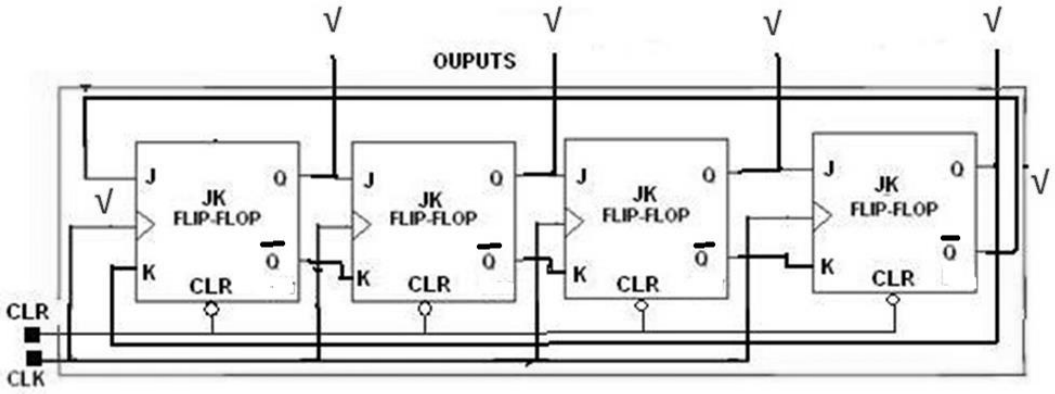
7.1



(10)

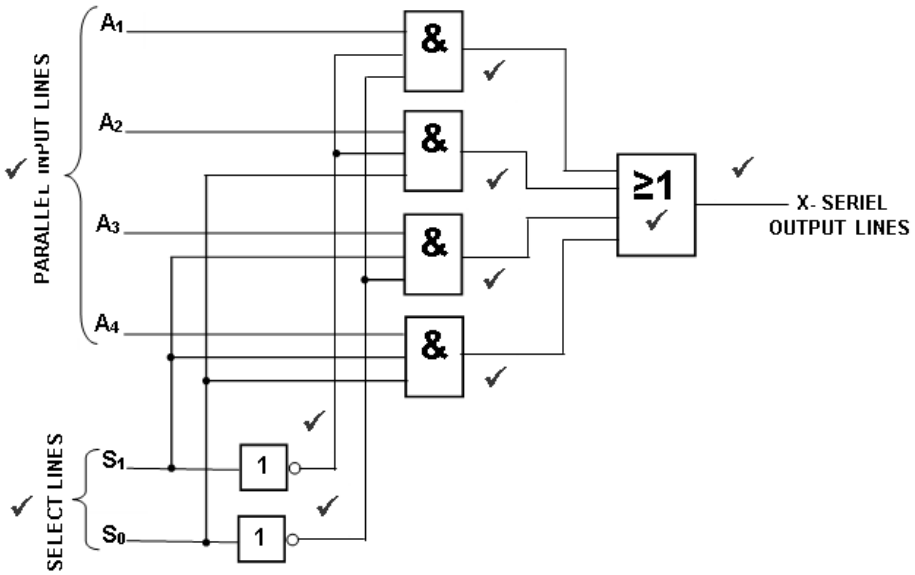
7.2 Both adders will have two inputs, A and B, as well as two outputs, namely sum and carry out. The difference is that a full adder has a carry in,✓✓ while a half adder does not have a carry in.✓✓ (2 + 2) (4)

7.3



(6)

7.4



(10)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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